CODESCHOOL · PROPOSAL FOR [SCHOOL NAME] · DRAFT

Offline Programme

Computational Thinking, coding and AI for Classes 3-8, delivered with printed workbooks, unplugged kits and teacher scripts. No student devices needed.

Prepared for: [Principal name], [School name], [City]

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Aligned to CBSE's CT&AI curriculum for Classes 3-8 (2026-27)

Contents

Draft

This is a draft for discussion with [School name]. Scope, timeline and prices are indicative and will be finalised after the pilot discussion. Nothing in this document is a commitment until both sides sign an agreement. A separate Online Programme proposal is available for schools with shared devices.

Covering letter . . . 4

1. Proposal at a glance . . . 4

2. About CodeSchool . . . 5

3. Understanding the school's needs . . . 5

- What CBSE asks schools to do from 2026-27 . . . 5
- What teachers still need . . . 5

4. Objectives and expected outcomes . . . 6

- Objectives . . . 6
- What students will be able to do by the end of the year . . . 6

5. Scope and deliverables . . . 6

- Included in the programme . . . 6
- Not included . . . 7

6. Programme design . . . 7

- Levels that follow the maths chapters . . . 7
- One routine for every lesson (about 40 minutes) . . . 7
- Coding without computers . . . 7
- Every child can take part . . . 7

7. Class-by-class overview . . . 8

8. Annual plan (session 2027-28) . . . 8

9. Sample weekly timetable . . . 9

10. How the offline programme runs . . . 9

11. Implementation plan . . . 9

- Phases . . . 9
- Pilot week by week . . . 10

12. Teacher support and professional development . . . 10

13. Assessment and reporting . . . 11

14. Parent engagement . . . 11

15. Governance, monitoring and quality . . . 11

16. What we need from the school . . . 12

17. Child safety and data protection . . . 12

18. Pilot success measures . . . 12
19. Investment . . . 13
Pilot terms . . . 13
After the pilot (indicative) . . . 13
How payment works . . . 13
20. Roles and responsibilities . . . 13
21. Risks and how we manage them . . . 14
22. Offline or online: choosing the right fit . . . 14
23. Frequently asked questions . . . 14
24. Assumptions and terms . . . 15
25. Glossary . . . 15
26. Next steps . . . 15
School response form . . . 15
Appendix A. Chapter-by-chapter curriculum map . . . 16
Class 3 (14 chapters, about 50 hours a year inside Maths and TWAU) . . . 16
Class 4 (14 chapters, about 50 hours a year) . . . 17
Class 5 (15 chapters, about 50 hours a year) . . . 18
Class 6 (10 CT chapters plus 4 AI chapters; 100 hours a year) . . . 18
Class 7 (8 CT chapters plus 4 AI chapters; 100 hours a year) . . . 19
Class 8 (7 CT chapters plus 4 AI chapters; 100 hours a year) . . . 19
Appendix B. AI strand for Classes 6-8 . . . 19
Class 6 . . . 19
Class 7 . . . 20
Class 8 . . . 20
Appendix C. Project kits for Classes 6-8 . . . 20
Standard 10-week project plan (about 2 hours a week) . . . 21
Appendix D. Assessment rubrics and report-card format . . . 21
Competency rubric for the observation journal . . . 21
Project rubric (used for every Class 6-8 project) . . . 21
Term summary for the report card (example) . . . 22
Appendix E. Sample lesson: "Robo makes choices" (Class 4, Level B3) . . . 22
Appendix F. Teacher development plan . . . 23

Covering letter

[Date]

Dear [Principal name],

Thank you for considering CodeSchool. From the 2026-27 session, CBSE asks every affiliated school to teach Computational Thinking and Artificial Intelligence (CT&AI) in Classes 3-8: 50 hours a year in Classes 3-5 and 100 hours a year in Classes 6-8, including 40 hours of projects. CBSE's free handbooks give your teachers a good set of puzzles, but they leave out several things teachers need in the classroom: coding activities, project materials, lesson scripts for the AI chapters, and a way to record and report progress.

This proposal describes our **Offline Programme**, which fills those gaps using printed workbooks, unplugged activity kits and ready-to-teach scripts. It needs no computers for students, works in any classroom, and is designed to be taught by your own maths, EVS and computer teachers within the hours CBSE has already allocated.

We propose to begin with a **free pilot** in two or three sections from January to March 2027, with clear success measures agreed in advance. If the pilot meets your expectations, the programme can run in all the classes you choose from April 2027.

We would welcome your feedback on this draft and would be glad to meet you and your coordinator to tailor it to [School name].

Warm regards,

Anoop Dobhal, CodeSchool · [Phone] · [Email]

1. Proposal at a glance

Item	Proposal
Programme	CodeSchool Offline Programme for Computational Thinking, coding and AI
Classes	3-8
What students use	A printed level workbook, unplugged activity kits, coding cut-outs and, in Classes 6-8, project kits and AI activity cards
What teachers get	Period-by-period scripts, answer keys, an observation journal, orientation and ongoing support
Devices needed	None for students. A classroom projector is useful but optional
Time	Within CBSE's existing allocation: about 1-2 periods a week in Classes 3-5 and 3-4 periods a week in Classes 6-8
Pilot	Free, 8-10 weeks, January to March 2027, in 2-3 sections (for example one Class 4 and one Class 7 section)
Full programme	From April 2027 (session 2027-28), in the classes the school chooses
Indicative price after the pilot	Rs 449 per student per year (Classes 3-5) and Rs 599 (Classes 6-8), on the school's book list
Student data	None. Only class-level pilot results, shared with the school's permission

2. About CodeSchool

CodeSchool is an Indian education start-up building a level-based programme that helps schools teach computational thinking, coding and AI in Classes 3-8. The programme is being designed specifically for CBSE's 2026-27 CT&AI curriculum and for real Indian classrooms, where many schools have few computers and teachers have little preparation time.

Our design is based on:

- a detailed review of **all twelve CBSE CT&AI student and teacher handbooks** for Classes 3-8 and CBSE's April 2026 webinar;
- international computer-science standards, including the **2026 CSTA PK-12 standards** and the **AI4K12 "Five Big Ideas in AI"**, which show the order in which children learn these ideas best;
- research on teaching programming, including the **PRIMM** lesson structure (predict, run, investigate, modify, make), **Parsons puzzles** and **unplugged** activities; and
- conversations with principals, coordinators and teachers at partner schools, which continue through the pilot.

3. Understanding the school's needs

What CBSE asks schools to do from 2026-27

	Classes 3-5	Classes 6-8
Hours per year	50, inside Mathematics and The World Around Us (TWAU)	100: 40 CT, 20 AI, 40 projects (two projects of 20 hours)
Who teaches	Maths and other subject teachers	Subject teachers for CT; the computer teacher for AI and projects
What CBSE provides free	Puzzle handbooks that follow the NCERT maths chapters, with one or two unplugged activities per class	The same, plus four AI chapters per class (concepts in the student book, answer keys in the teacher book)
Assessment CBSE expects	Written tests with CT puzzles, group activities, a Teacher Observation Journal	Written tests, practical examinations, group activities, projects, a Teacher Observation Journal

What teachers still need

Gap in the free handbooks	What the Offline Programme adds
No coding: students follow algorithms but never write one	"Now you build it" coding on paper for every chapter
Hard puzzles with no hints or easier steps	Levels with warm-ups, hint cards and challenge levels
No project briefs, rubrics or examples for the 40 project hours	Two complete project kits per class with rubrics
AI teacher books contain answer keys, not lesson scripts	A teacher script and unplugged activity for every AI chapter
Hands-on AI only in Class 8, and only with computers and webcams	Unplugged AI card labs in Classes 6-8, plus a projector demonstration
A Teacher Observation Journal with no template or report-card format	A competency journal and a term summary ready for the report card

> **Our approach:** we do not replace CBSE's free books or the NCERT maths textbook. We sit on top of them, chapter by chapter, and add the missing pieces, so teachers use one joined-up plan.

4. Objectives and expected outcomes

Objectives

1. Help teachers deliver CBSE's CT&AI hours with confidence, using ready scripts that fit one period.
2. Move students from **following** algorithms to **building** them, starting in Class 3.
3. Give Classes 6-8 complete, assessable project kits for their 40 project hours.
4. Make AI understandable through hands-on, unplugged activities in every year from Class 6.
5. Give the coordinator a simple, competency-based record for report cards.

What students will be able to do by the end of the year

Classes	Expected outcomes
3	Give and follow precise step-by-step instructions; use repeat (loops) and simple events; find and fix a mistake in a sequence
4	Use if-then-else choices and loops inside loops; trace what a set of instructions will do; explain their algorithm in words
5	Use variables and inputs on paper; compare two ways of solving a problem; design a simple game with rules
6	Create reusable steps (functions); organise data in lists and charts; explain how AI learns from examples; complete two projects
7	Read and predict simple programs; use data to make predictions; explain classification, clustering and bias; complete two projects
8	Read and trace simple Python; follow the AI project cycle; audit an AI system for fairness; complete two projects

5. Scope and deliverables

Included in the programme

Component	Description	Who receives it	Classes
Level workbook	Short levels for each NCERT maths chapter, from warm-ups with hints to challenges	Each student	3-8
"Now you build it" coding	Paper code cards and cut-out blocks: students write instructions, and classmates "run" them	Each student (in workbook)	3-8
Unplugged activity kits	Printable games, cards and floor grids, with a script for each	Each class section	3-8
Project kits	Two 20-hour projects a year: brief, data sheets, worked example, rubric, presentation template	Each student; teacher guide for the teacher	6-8
AI activity cards	Unplugged classification, data and bias activities for each AI chapter	Each class section	6-8
Teacher script book	Period-by-period plans with timings, answers and common mistakes	Each teacher	3-8
Observation journal	Printable competency checklist per class and a term summary	Each teacher	3-8
Orientation and support	Orientation at school, workshops, weekly check-ins during the pilot, question channel and short videos	All participating teachers	All

Not included

- Hardware, devices, internet or lab set-up.
- Replacing CBSE's handbooks or the NCERT maths textbook.
- Board-examination preparation for Classes 9-12.
- Home tuition or any parent-paid add-on during the pilot.

6. Programme design

Levels that follow the maths chapters

Every lesson is tied to the NCERT maths chapter the class is studying, so teachers find "today's" material in one place. Within each chapter, students move through short levels: a **warm-up** (with hints), a **guided level**, a **build level** (where they create their own instructions) and a **challenge**. A student who finishes early moves to the challenge; a student who struggles uses a hint card. The teacher keeps the whole class on the same chapter. The full chapter-by-chapter map is in **Appendix A**.

One routine for every lesson (about 40 minutes)

Step	What happens	Minutes
1. Predict and run	Students predict what a short set of instructions will do, then act it out	8
2. Investigate	Pairs answer tracing questions about why it worked	8
3. Modify	Students change the instructions to do something new	8
4. Parsons puzzle	Students put jumbled instruction cards in the right order	8
5. Make	Pairs create their own rule or program and swap with another pair	6-8

A complete sample lesson is in **Appendix E**.

Coding without computers

Students learn real programming ideas (sequences, loops, events, conditions, variables and functions) using paper code cards and cut-out blocks that look like the blocks used in Scratch-style coding. Classmates or the teacher "run" the program by following it exactly, which makes mistakes visible and fun to fix. In Class 8, students read and trace short programs written in Python on paper. If a projector is available, the teacher can show the same program on screen.

Every child can take part

- **Different paces:** warm-up levels and hint cards for students who need support; challenge levels for those ready to go further.
- **Language:** teacher scripts and key student pages can be provided in Hindi as well as English (to be confirmed for the school's medium of instruction).
- **Collaboration:** most activities are done in pairs or small groups, so students who find reading or writing difficult can contribute through discussion and acting out.
- **Inclusion:** physical and visual activities (floor grids, cards) help students with different learning needs; teachers can adapt timings.

7. Class-by-class overview

Class	Computational thinking and coding	AI	Projects
3	Sequences, repeat loops and events on floor grids; secret codes (ciphers); path puzzles	Sorting objects by features (enrichment)	Short class challenges
4	If-then-else choices ("Robo makes choices"); loops inside loops; error-finding with parity cards	Spotting patterns in pictures (enrichment)	Short class challenges
5	Variables and inputs on paper; searching and sorting strategies; a capstone game designed on paper	How examples teach a machine (enrichment)	Short class challenges
6	Reusable steps (functions); lists and data from the Data Handling chapter	What AI is; classifying with picture cards; data and patterns; digital ethics	Weather across states; a smarter school routine
7	Data tables and charts; prediction from patterns; reading simple programs	Classification, clustering, AI in industries, charts and bias	Nutri-Quest food planner; sort it with AI
8	Reading and tracing simple Python on paper; debugging	The AI project cycle; applications; fairness; responsible AI	Sports participation data; fair AI audit

Detailed maps: chapters in **Appendix A**, AI chapters in **Appendix B**, projects in **Appendix C**.

8. Annual plan (session 2027-28)

The programme fits inside CBSE's allocation. A typical school year of about 32 teaching weeks is planned as follows; the school can move units between terms.

	Term 1 (April-September)	Term 2 (October-March)	Total
Classes 3-5: CT and coding within Maths/TWAU	About 26 hours: first half of the maths chapters	About 24 hours: second half of the chapters plus the capstone	50 hours
Classes 6-8: CT and coding	About 20 hours	About 20 hours	40 hours
Classes 6-8: AI	About 10 hours (AI chapters 1-2)	About 10 hours (AI chapters 3-4)	20 hours
Classes 6-8: Projects	Project 1: about 20 hours	Project 2: about 20 hours	40 hours

9. Sample weekly timetable

Class	Periods per week (40 minutes)	Example
3-5	1-2, taken from Maths or TWAU periods	Tuesday maths period: CT level for the current chapter; Friday (alternate weeks): unplugged activity
6-8	3-4	Monday: CT level (maths teacher); Wednesday: AI activity (computer teacher); Thursday and Friday: project work (computer teacher)

10. How the offline programme runs

Element	Classes 3-5	Classes 6-8
Room	Normal classroom; floor space or corridor for grid games	Normal classroom; projector optional for demonstrations
Materials per student	Level workbook with coding cut-outs	Level workbook, project booklet
Materials per section	Activity kit box: cards, floor-grid template, cipher wheels, sorting cards	Activity kit box: AI cards, data sheets, presentation templates
Materials per teacher	Script book, journal, answer keys	Script book, project teacher guides, journal, rubrics
Preparation time	About 5-10 minutes per lesson using the script	About 10 minutes per lesson; projects have a week-by-week plan
Replacing materials	Printable masters supplied for any lost or extra copies	Same

11. Implementation plan

Phases

Phase	When	What happens	Output
1. Agree scope	December 2026	Meet principal and coordinator; choose pilot sections and teachers; share specimen materials	Signed pilot letter
2. Prepare	First week of January 2027	Deliver pilot materials; 2-hour teacher orientation at school; fix the timetable	Teachers ready; timetable fixed
3. Pilot	January to mid-March 2027 (8-10 weeks)	Teachers run lessons; weekly check-ins; two workshops; the coordinator keeps the journal	Lessons delivered; journal entries
4. Review	Mid-March 2027	Compare results with the success measures; teacher and student feedback	Pilot report for the school
5. Roll out	April 2027 onwards	Book-list entry; orientation for all teachers; materials for all chosen sections; termly workshops	Programme in all chosen classes

Pilot week by week

Week	Classes 3-5 pilot section	Classes 6-8 pilot section	CodeSchool support
0	Orientation; baseline puzzle check (15 minutes)	Orientation; baseline puzzle check	Orientation at school
1-2	Levels for the current maths chapter (sequences and loops)	CT levels; AI chapter activity; project launch (define the problem)	Weekly check-in
3-4	Conditionals ("Robo makes choices")	AI card lab; project data collection	Mid-pilot workshop (week 4)
5-6	Next chapter's levels; unplugged kit activity	CT levels; project analysis and building	Weekly check-in
7-8	Build levels and challenges	AI bias activity; project testing and improvement	Assessment workshop (week 8)
9	Capstone mini-challenge	Project reflection	Weekly check-in
10	End puzzle check; journal summary	Project presentations; end puzzle check; journal summary	Review meeting and pilot report

Book-list timing: most schools finalise next year's book list in December-January, before the pilot ends. Schools that want the programme from April 2027 can review specimen copies and a sample lesson in December and decide then; the pilot will still shape how the programme is taught.

12. Teacher support and professional development

- **Orientation (2 hours, at school)** before the pilot, with a live demonstration lesson.
- **Scripts for every lesson:** timings, what to say, answers and the mistakes students usually make.
- **Weekly check-ins** (15 minutes) during the pilot, plus a shared question channel.
- **Two pilot workshops** (60 minutes each) and **one workshop per term** in the full year.
- **Short how-to videos** for each activity.
- **Certificate of completion** for teachers who complete the orientation and termly workshops.
- **No specialist needed:** the programme is designed so that a maths or EVS teacher can run it.

The full plan and orientation agenda are in **Appendix F**.

13. Assessment and reporting

- **Level checks** at the end of each chapter's levels, marked with an answer key in minutes.
- **Observation journal** with a four-level competency rubric (beginning, developing, secure, extending) for abstraction, pattern recognition, decomposition and algorithmic thinking, plus AI literacy in Classes 6-8.
- **Project rubrics** with six criteria, including testing and fairness.
- **Term summary** for the report card, with a grade and comment per area.
- **Baseline and end puzzle checks** (15 minutes each) to show progress over the pilot and the year.

Rubrics and an example report-card summary are in **Appendix D**.

14. Parent engagement

- **Information letter** to parents of pilot sections explaining what CT&AI is and what their child will do (provided by CodeSchool, sent by the school).
- **Take-home challenges:** an optional puzzle page each month that children can solve with their family, with no device needed.
- **Showcase:** at the end of each term, students present projects or games to parents (Classes 6-8) or demonstrate a "Robo" activity (Classes 3-5).
- **No compulsory purchases:** parents can buy the workbook through the school's book list or from booksellers and online.

15. Governance, monitoring and quality

Mechanism	Frequency	Who	Purpose
Pilot lead check-in	Weekly during the pilot	Coordinator and CodeSchool	Progress, issues, next steps
Progress note	Monthly	CodeSchool to principal	Lessons delivered, level checks, teacher feedback
Review meeting	End of pilot and end of each term	Principal, coordinator, CodeSchool	Results against success measures; changes needed
Classroom visit	Once during the pilot, with permission	CodeSchool	Support teachers and improve materials
Escalation	As needed	Coordinator to CodeSchool founder	Any concern is answered within two working days

16. What we need from the school

Need	Detail
A pilot lead	The coordinator or a senior teacher as our single point of contact
Teachers	One named teacher per pilot section, available for the orientation and two workshops
Timetable	The agreed periods per week for the pilot sections
Space	A classroom with some floor space for grid activities
Feedback	Two short feedback conversations (mid-pilot and end) and access to journal summaries (class level)
Communication	Sending the CodeSchool information letter to parents of pilot sections

17. Child safety and data protection

- **No student accounts and no student data.** The offline programme works entirely on paper.
- **No photos or recordings of students** are taken by CodeSchool. Any showcase photos are taken and kept by the school under its own policy.
- **Pilot results are shared only as class-level summaries** (for example, the share of students who passed a level check), with the school's permission.
- **Student names stay in the school's journal and records.** If any student data is ever needed in future, it will be handled under India's Digital Personal Data Protection Act, 2023, with the school in control and a written agreement in place.

18. Pilot success measures

These are agreed in advance so the pilot result is clear to both sides. Targets are suggestions and can be changed.

Area	Measure	Suggested target
Teacher delivery	Planned pilot periods the teacher ran without outside help	80%
Fit	Lessons that fit within one period	90%
Learning	Students passing the level check after each chapter	70%
Progress	Average improvement from baseline to end puzzle check	A clear gain, reported by class
Engagement	Students completing the "make" step	75%
Teacher time	Preparation time per lesson	10 minutes or less
Teacher view	Would use it again next session (1-5)	4 or more
Reporting	Journal and term summary usable for report cards	Yes

19. Investment

Pilot terms

- The pilot is **free** for 2-3 sections: materials, orientation, workshops and weekly support.
- There is **no obligation** to buy afterwards.
- Pilot materials remain with the school for the pilot students to keep.

After the pilot (indicative)

Package	What is included	Indicative price
Classes 3-5 Practice Pack	Level workbook with coding cut-outs, take-home challenges	Rs 449 per student per year (MRP)
Classes 6-8 Projects and AI Pack	Level workbook, two project booklets, take-home challenges	Rs 599 per student per year (MRP)
Section kits and teacher materials	Activity kit boxes, script books, journals, rubrics, orientation, termly workshops and support	Included with the packs

How payment works

- The packs go on the school's **published book list**, and can **replace the existing Computers book**, so the total cost to parents stays about the same.
- Parents can buy the packs through the school's usual bookseller or from other booksellers and online. **No parent is required to buy from one source**, and **no new fee head** is needed.
- If the school prefers to buy the packs centrally from its own budget, CodeSchool can invoice the school directly.
- Prices are indicative, will be confirmed after the pilot, and will be held for the 2027-28 session once confirmed. Taxes, if any, will follow the final package structure.

20. Roles and responsibilities

Task	CodeSchool	School
Lesson materials, kits and scripts	Provides and updates	Reviews and gives feedback
Curriculum mapping to the school's chapter order	Prepares	Confirms
Teacher orientation, workshops and support	Leads	Releases teachers to attend
Timetabling	Advises	Decides
Teaching the lessons	Supports	Delivers
Observation journal and term summary	Provides formats	Keeps records and reports
Parent communication	Drafts letters	Sends to parents
Pilot review and report	Prepares	Reviews and decides

21. Risks and how we manage them

Risk	How we manage it
Teachers have little time to prepare	Every lesson has a 5-10 minute script; the orientation covers the routine once
Lessons run over a period	Every lesson is designed for 40 minutes, with an optional extension
Puzzles too hard for some students	Hint cards and warm-up levels in every chapter
A teacher leaves or is absent	Scripts let another teacher take over; CodeSchool runs a catch-up orientation
CBSE publishes its own project list or rubrics	Project kits will be realigned to CBSE's topics as soon as they are published
Pilot classes miss periods for events or exams	The plan allows 8-10 weeks and can pause and resume
Materials are lost	Printable masters are supplied for replacements

22. Offline or online: choosing the right fit

Question	Offline Programme	Online Programme
Student devices needed?	No	Yes, shared devices (about 1 per 2 students during the lesson)
Where students practise	Paper workbook and classroom activities	App or browser, with automatic checking
Coding	On paper, with cut-out blocks	Block coding on screen
AI	Unplugged card labs; projector demonstration	Hands-on model training with supplied datasets
Teacher reporting	Printable journal	Live class dashboard and report-card export
Student data	None	School-managed accounts using roll numbers

Many schools may prefer a **hybrid**: the Offline Programme in Classes 3-5 and the Online Programme in Classes 6-8, where the computer lab is available. A separate Online Programme proposal is available.

23. Frequently asked questions

Does this replace CBSE's free handbooks? No. It works alongside them, chapter by chapter, and adds what they do not include.

Do we need a computer lab? No. Every lesson works on paper. A projector helps for some demonstrations but is optional.

Who teaches it? Your own maths and EVS teachers in Classes 3-5, and your subject and computer teachers in Classes 6-8, as CBSE suggests. No specialist is needed.

Will it take extra periods? No. It uses the hours CBSE has already allocated to CT&AI.

Is coding really possible without computers? Yes. Students learn the same ideas (sequences, loops, conditions, variables) by writing and "running" programs on paper, which research shows is an effective start. Schools can add screen-based coding later through the Online Programme.

How is it assessed? Through level checks, the observation journal, project rubrics and a term summary for the report card (Appendix D).

What if CBSE changes the curriculum? The programme is built around skills, with CBSE mapping kept separate, so it can be updated quickly.

Can we stop after the pilot? Yes. There is no obligation to continue.

24. Assumptions and terms

- This proposal assumes the school follows CBSE's CT&AI curriculum and the NCERT mathematics textbooks for Classes 3-8.
- Pilot dates assume a signed pilot letter by mid-December 2026.
- Prices are indicative until confirmed in writing after the pilot.
- CodeSchool retains ownership of its materials; the school may use them for its own students during the agreement.
- Formal terms, including any data-related terms, will be set out in a separate agreement.
- "CodeSchool" is the programme's working name.

25. Glossary

Term	Meaning
CT&AI	Computational Thinking and Artificial Intelligence, CBSE's subject for Classes 3-8 from 2026-27
Computational thinking	Solving problems in steps a person or computer can follow: abstraction, pattern recognition, decomposition and algorithms
Algorithm	A precise set of steps to solve a problem
Unplugged activity	Learning a computing idea through games, cards or movement, without a computer
Block coding	Programming by snapping together visual blocks (here, paper cut-out blocks) instead of typing
Loop	An instruction to repeat steps
Conditional	An if-then-else choice in a program
Variable	A named store for a value that can change, such as a score
Function	A named set of steps that can be reused
Parsons puzzle	Putting jumbled lines of code in the right order
PRIMM	A lesson routine: predict, run, investigate, modify, make
Classifier	An AI model that sorts examples into groups it has learned
Bias (in AI)	Unfair results caused by unbalanced or incomplete data

26. Next steps

1. A short meeting with the principal and coordinator to agree the pilot sections and teachers.
2. CodeSchool shares specimen materials and the sample lesson.
3. The school returns the response form below and signs a one-page pilot letter.
4. Materials delivery, teacher orientation and the pilot start in January 2027.

School response form

Fill this in on screen or on paper and return it to CodeSchool. It is an expression of interest, not a contract.

School name	
Board and city	
Contact person and role	
Phone / email	
Classes and sections for the pilot	
Approximate number of students	

Preferred start date	
Programme of interest	-
Questions or changes you would like	

Appendix A. Chapter-by-chapter curriculum map

CBSE's CT&AI handbooks follow the NCERT mathematics textbook chapter by chapter, so CodeSchool does the same. For every chapter, the map below shows the main computational-thinking (CT) skill CBSE's handbook practises and the CodeSchool level or activity that adds a "now you build it" step. **This is the planned mapping for the pilot; it will be confirmed with the school's teachers and adjusted if the school follows a different chapter order.**

CodeSchool levels are grouped in bands. **Band B (Builders)** covers Classes 3-5: B1 sequences and loops, B2 events and patterns, B3 conditionals (if-then-else), B4 nested and repeat-until loops, B5 variables and inputs, B6 CT puzzle strategies, B7 capstone game. **Band C (Makers)** covers Classes 6-8: C1 functions, C2 lists and data, C3 AI literacy I, C4 the code behind the blocks, C5 reading code, C6 writing simple code, C7 AI literacy II, C8 projects.

Class 3 (14 chapters, about 50 hours a year inside Maths and TWAU)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. What's in a Name?	Patterns; step-by-step encoding (Caesar cipher activity)	B1 "Secret messages": write and follow encoding steps
2. Toy Joy	Abstraction: views of 3D objects	B1 "Build from instructions": block towers from sequence cards
3. Double Century	Number sequences	B1 "Count-along loops": the repeat card
4. Vacation with My Nani Maa	Decomposition of multi-step problems	B1 "Plan the trip": order and debug a sequence
5. Fun with Shapes	Shape patterns, folding	B2 "Shape patterns with repeat"
6. House of Hundreds I	Place-value logic	B2 "Robot counter": events ("when tapped, add 10")
7. Raksha Bandhan	Equal groups, data sufficiency	B2 "Repeat the group"
8. Fair Share	Halves and fair rules	B2 "Split it fairly": sorting by a rule
9. House of Hundreds II	Clue-based reasoning	B2 "Clue detectives": which clues are enough?
10. Fun at Class Party!	Ordering by measurement	B2 "Line up by rule" (sorting)
11. Filling and Lifting	Rule with a branch ("on a white block skip two, on a black block skip one"); weighing logic	B3 preview "Skip rules": first if-then cards
12. Give and Take	Step-by-step exchanges	B2 "Trace the exchange": keeping count (variables preview)
13. Time Goes On	Sequences in time	B2 "Clock loops"
14. The Surajkund Fair	Paths, grids, pathway tiles	B1-B2 "Robot routes": path tiles and grid programs

Class 4 (14 chapters, about 50 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Shapes Around Us	Views, faces, edges	B1 review: "Build from instructions"
2. Hide and Seek	Paths and turns; a rule with an exception	B3 "Robo makes choices" (sample lesson, Appendix E)
3. Patterns Around Us	Parity and error detection (Coin Flip Magic activity)	B2 "Find the flipped coin": checking rules
4. Thousands Around Us	Ordering (floor Sorting Network activity)	B4 "Sorting machines": compare-and-swap steps
5. Sharing and Measuring	Fair sharing, fractions	B3 "If it is even, then…"
6. Measuring Length	Shortest paths on tiles	B4 "Shortest route": loops with turns
7. The Cleanest Village	Tracing a button-press sequence	B3 "Button machine": trace and predict
8. Weigh It, Pour It	Comparing quantities	B3 "Heavier or lighter?": if-else decisions
9. Equal Groups	Repeated groups	B4 "Loops inside loops"
10. Elephants, Tigers and Leopards	Data and clues	B2 "Clue tables"
11. Fun with Symmetry	Reflections and turns	B4 "Mirror drawings": repeat with turns
12. Ticking Clocks and Turning Calendar	Cycles and counting	B4 "Repeat until": day and time cycles
13. The Transport Museum	Classifying vehicles by features	B3 "Sort by rules": decision cards
14. Data Handling	Tables and tallies	B2 "Count and chart"

Class 5 (15 chapters, about 50 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. We the Travellers I	Multi-step journeys	B4 review: routes with loops and turns
2. Fractions	Part-whole patterns	B5 "Keep score": variables that change
3. Angles as Turns	Turns and directions	B4 "Turtle turns": draw shapes with angles
4. We the Travellers II	Planning with constraints	B5 "Fuel left?": inputs and variables
5. Far and Near	Measuring and comparing	B5 "Compare and decide"
6. The Dairy Farm	Large quantities, grouping	B5 "Counters and totals"
7. Shapes and Patterns	Growing patterns	B4 "Pattern generators"
8. Weight and Capacity	Searching with a weighing scale (linear vs divide-and-conquer)	B6 "Search strategies": which search is faster?
9. Coconut Farm	Multiplication patterns	B5 "Harvest calculator"
10. Symmetrical Designs	Reflection and repetition	B4 "Design machines"
11. Grandmother's Quilt	Tiling patterns	B6 "Tile puzzles": decomposition
12. Racing Seconds	Timing and ordering	B5 "Stopwatch": variables over time
13. Animal Jumps	Skip patterns on number lines	B6 "Jump rules": loops with conditions
14. Maps and Locations	Grids and directions	B7 capstone "Treasure map game" (design)
15. Data Through Pictures	Pictographs and tables	B7 capstone: add a score and finish the game

Class 6 (10 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Patterns in Mathematics	Number and shape patterns	C1 "Make a pattern block": functions
2. Lines and Angles	Angles, constructions	C1 "Draw with functions": reusable steps
3. Number Play	Number rules and puzzles	C1 "Number machines"
4. Data Handling and Presentation	Tables, charts	C2 "Lists and charts"
5. Prime Time	Factors; 12-point circle step-rule activity	C2 "Prime finder": loops over lists
6. Perimeter and Area	Measurement rules	C1 "Area calculator" with inputs
7. Fractions	Visual patterns of fractions	C2 "Fraction lists"
8. Playing with Constructions	Step-by-step constructions	C1 "Construction scripts"
9. Symmetry	Folding and transformation	C1 "Symmetry painter"
10. The Other Side of Zero	Negative numbers, number lines	C2 "Temperature log": data with negatives

Class 7 (8 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. Large Numbers Around Us	Place value at scale	C2 "Big data lists"
2. Arithmetic Expressions	Order of operations	C4 "The code behind the blocks": expressions
3. A Peek Beyond the Point	Decimals; punched-card binary sorting activity	C4 "Binary and sorting"
4. Expressions using Letter-Numbers	Variables and algebra	C4-C5 "Variables in code"
5. Parallel and Intersecting Lines	Geometry rules	C5 "Read and predict": tracing programs
6. Number Play	Logic puzzle punch cards (true/false, if-then)	C5 "Logic and conditions"
7. A Tale of Three Intersecting Lines	Triangle rules	C5 "Debug the program"
8. Working with Fractions	Fraction operations	C4 "Fraction functions"

Class 8 (7 CT chapters plus 4 AI chapters; 100 hours a year)

NCERT chapter	CT focus in CBSE handbook	CodeSchool level or activity
1. A Square and a Cube	Powers and patterns	C5-C6 "Squares and cubes" in simple code
2. Power Play	Exponents	C6 "Repeat and multiply": loops in code
3. A Story of Numbers	Number systems; binary and ternary "pipes" activity	C6 "Number system converter"
4. Quadrilaterals	Classification by properties	C6 "Shape classifier" with if-else
5. Number Play	Logic and rules	C6 "Rule checker"
6. We Distribute Yet Things Multiply	Distribution and expressions	C6 "Expression tester"
7. Proportional Reasoning	Ratios and scaling	C6 "Recipe scaler" with variables

Appendix B. AI strand for Classes 6-8

CBSE's 20 AI hours a year in Classes 6-8 are organised into four chapters per class. CBSE's student books explain the concepts; the teacher books give exercises and answer keys. CodeSchool adds a teacher script and a hands-on activity for every chapter, in an offline form (cards, paper and one projector) and an online form (browser activities with supplied datasets). Schools can use either form.

Class 6

CBSE AI chapter	What students learn	Offline activity	Online activity
1. Introduction to AI and everyday examples	What AI is; AI vs automation; supervised, unsupervised and reinforcement learning	"Is it AI?" sorting cards; "human classifier" game with labelled pictures	Train a picture sorter with a supplied set of labelled images
2. Basic data concepts	Types of data; collecting and organising data	Class survey on paper; build a frequency table	Enter survey data and see charts update
3. Simple pattern recognition and decision making	Finding patterns; observations vs conclusions	Pattern cards and decision trees on chart paper	Build a decision tree and test it on new examples
4. Ethics and digital responsibility	Passwords, phishing, digital footprint, plagiarism	Scenario cards and class debate	Interactive scenarios with instant feedback

Class 7

CBSE AI chapter	What students learn	Offline activity	Online activity
1. AI domains and applications	Data science, computer vision, language; classification, regression, clustering	Card labs: classify fruits by features; group "unlabelled" cards into clusters	Train a classifier, a simple prediction line and a clustering tool on small datasets
2. AI in industries	Case studies of AI in health, transport and farming	Case-study cards and role-play	Explore case studies with short quizzes
3. Data visualisation and analysis	Bar, line and pie charts; reading trends	Hand-drawn charts from class data	Build charts from a dataset and spot trends
4. Ethics and AI bias awareness	How unbalanced data causes unfair results	"Lopsided deck" activity: a biased card set gives unfair answers	Retrain a model after rebalancing its data and compare results

Class 8

CBSE AI chapter	What students learn	Offline activity	Online activity
1. AI project lifecycle	Define the problem, collect data, test, reflect and improve	Plan an AI project on a lifecycle worksheet	Run the full lifecycle with a no-code model
2. AI and its applications	No-code AI tools; environment, health and education examples	Teacher-led demonstration of CBSE's waste-classifier lab on one projector	Guided version of CBSE's waste-classifier and voice-command labs, with supplied data
3. Data and fairness in AI	Bias in datasets; rebalancing	Card-based fairness audit	Guided version of CBSE's cat/dog bias lab: test, rebalance, retrain
4. Ethics and responsible AI	Privacy, misinformation, responsible use	Debate and "spot the fake" cards	Interactive misinformation and privacy scenarios

Appendix C. Project kits for Classes 6-8

CBSE allots **40 hours a year to two required projects of 20 hours each** in Classes 6-8, assessed by the computer teacher. CBSE has said a project list with rubrics is coming; until it is published, CodeSchool's kits follow the topics CBSE presented in its April 2026 webinar, and will be realigned to CBSE's official list as soon as it appears.

Every kit contains: a student brief, a teacher guide with a week-by-week plan, data sheets or datasets, a worked example, a marking rubric (Appendix D), and a presentation template.

Class	Project 1 (20 hours)	Project 2 (20 hours)
6	**Weather across states:** collect and compare weather data from different states, find patterns and present conclusions	**A smarter school routine:** design an algorithm and flowchart to solve a school problem (for example, library returns or lunch queues), then test it
7	**Nutri-Quest weekly food planner:** use nutrition data and rules to design a balanced week of meals	**Sort it with AI:** build a classifier for school waste (cards offline, a trained model online) and measure its accuracy
8	**Sports participation and performance:** collect sports data, analyse it and recommend changes	**Fair AI audit:** test a classifier for bias, rebalance the data and present recommendations

Standard 10-week project plan (about 2 hours a week)

Weeks	Stage	What students do	Evidence for assessment
1-2	Define	Understand the problem, form teams, plan questions	Problem statement and plan
3-4	Collect	Gather data or design the first version of the algorithm	Data sheet or first flowchart
5-6	Build	Organise data, make charts, build the algorithm or model	Charts, flowchart or model
7-8	Test and improve	Test with new cases, find errors, improve	Test log with changes made
9	Reflect	Discuss fairness, limits and what was learned	Short reflection
10	Present	Share findings with the class or at a showcase	Presentation and final product

Appendix D. Assessment rubrics and report-card format

CBSE asks for continuous, competency-based assessment for CT&AI, including written tests with CT puzzles, group activities, practical work, projects and a **Teacher Observation Journal**, but it has not published a template or report-card format. CodeSchool provides the following, mapped to CBSE's competency areas.

Competency rubric for the observation journal

Competency	Beginning	Developing	Secure	Extending
Abstraction	Needs help to ignore unimportant details	Picks out key details with prompts	Identifies the key details on their own	Explains which details matter and why
Pattern recognition	Spots simple repeating patterns with help	Continues patterns with one change	Finds and extends patterns with two or more changes	Creates and explains new patterns
Decomposition	Needs help to break a problem into steps	Breaks a familiar problem into parts	Breaks new problems into clear parts	Plans the parts and their order independently
Algorithmic thinking	Follows a short sequence with help	Follows and fixes simple sequences	Writes sequences with loops and conditions	Designs, tests and improves their own algorithms
AI literacy (Classes 6-8)	Names everyday uses of AI	Explains that AI learns from data	Trains or tests a model and explains its results	Identifies bias and suggests fairer data

Project rubric (used for every Class 6-8 project)

Criterion	1 Beginning	2 Developing	3 Secure	4 Extending
Problem definition	Problem unclear	Problem stated but vague	Clear problem and goal	Clear problem, goal and success measure
Data or algorithm	Little data or no clear steps	Some data or steps, with gaps	Complete data or working algorithm	Well-organised data or efficient, tested algorithm
Testing and improvement	Not tested	Tested once	Tested with new cases and improved	Systematic testing with recorded improvements
Ethics and fairness	Not considered	Mentioned	Discussed with an example	Changed the work to make it fairer
Communication	Hard to follow	Partly clear	Clear and organised	Clear, engaging and well supported by evidence
Teamwork	Little contribution	Some contribution	Shared work fairly	Led and supported others

Term summary for the report card (example)

Area	Grade	Teacher comment
Computational thinking (overall)	Secure	Breaks problems into steps and writes loops confidently
Coding (build levels completed)	12 of 15 levels	Ready for conditionals with variables
AI literacy (Classes 6-8)	Developing	Trained a classifier; working on explaining bias
Project	3 (Secure)	Clear charts; testing could be more systematic

Appendix E. Sample lesson: "Robo makes choices" (Class 4, Level B3)

This lesson pairs with the Class 4 NCERT chapter *Hide and Seek* and teaches if-then-else choices. It runs in one 40-minute period with no devices (a projector is optional). A full printable version with the student sheet is available on request.

Step	Minutes	Teacher does	Students do
1. Predict and run	8	Draws a row of five squares (START, PUDDLE, plain, PUDDLE, plain) and shows a program: repeat 4 times, move forward, *if* on a puddle *then* say "Splash!" *else* say "Step"	Write what Robo will say, then one student acts as Robo while the class reads the program aloud (answer: Splash, Step, Splash, Step)
2. Investigate	8	Asks four tracing questions, including "what if the if-else were outside the loop?"	Discuss in pairs and explain why the check happens every time
3. Modify	8	Draws a new row with STAR squares	Change only the if-else so Robo says "Star!" on star squares
4. Parsons puzzle	8	Hands out jumbled blocks, one of which is a trick	Put the blocks in order so Robo reaches the goal without hitting a wall
5. Make and exit ticket	8	Invites pairs to invent their own rule	Draw five squares, write "If Robo is on a ___ square, then ___, else ___", swap and predict; answer a two-question exit ticket

Common mistakes the script prepares teachers for: thinking both branches run; thinking the "if" is checked only once.

Appendix F. Teacher development plan

When	Session	Length	Content
Before the pilot	Orientation at school	2 hours	Why CT&AI; the five-step lesson routine; a live demonstration lesson; the observation journal; questions
Pilot weeks 1-10	Weekly check-in	15 minutes	What worked, what did not, next week's lessons
Pilot week 4	Mid-pilot workshop	60 minutes	Teaching conditionals and loops; managing different paces; project start-up (Classes 6-8)
Pilot week 8	Assessment workshop	60 minutes	Using the rubrics and journal; preparing the term summary
Full year	Termly workshop	90 minutes per term	New units, AI chapters, project kits, sharing practice across teachers
Ongoing	Question channel and short videos	As needed	Quick help and how-to clips for each activity

Teachers who complete the orientation and the termly workshops receive a CodeSchool certificate of completion.

Orientation agenda (2 hours): welcome and goals (10 minutes); CBSE CT&AI in one page (15); the lesson routine, experienced as students (35); planning one lesson from the script (20); the journal and rubrics (20); timetable and support (10); questions (10).

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